

MODEL RESEARCH DOCUMENT

Project Title

AI-Powered Workforce Analytics & Talent Intelligence Dashboard

Introduction

This document presents additional machine learning, deep learning, ensemble, clustering, anomaly detection, and generative AI models that can enhance the AI-Powered Workforce Analytics & Talent Intelligence Dashboard. These models support predictive workforce analytics, employee performance analysis, talent intelligence, skill-gap analysis, workforce segmentation, anomaly detection, and AI-assisted reporting. The comparison below highlights their primary use cases, advantages, and expected performance for structured enterprise workforce datasets.

Additional Models and Technologies

Model	Use Case	Advantages	Expected Performance
Extra Trees	Employee risk prediction	Fast ensemble, low variance	92-96%
HistGradient Boosting	Large workforce datasets	Efficient boosting	93-96%
TabNet	Tabular HR analytics	Interpretable deep learning	94-97%
FT-Transformer	Structured workforce data	High tabular accuracy	95-98%
TabTransformer	Mixed categorical features	Learns feature interactions	94-98%
Voting Classifier	Model aggregation	Stable predictions	93-97%
Stacking Classifier	Meta ensemble	Excellent generalization	95-98%
Isolation Forest	Fraud/outlier detection	Detects anomalies	High Precision
Local Outlier Factor	Behavior analysis	Density-based detection	High Precision
One-Class SVM	Novelty detection	Unlabeled anomaly detection	High Precision
DBSCAN	Employee segmentation	Finds irregular clusters	Good
Gaussian Mixture	Soft clustering	Probabilistic grouping	Good
UMAP	Visualization	Fast dimensionality reduction	Excellent
Autoencoder	Feature learning	Learns hidden patterns	High
RoBERTa	Resume & text analysis	Improved NLP accuracy	F1 92-95%
DeBERTa	Skill extraction	Strong contextual embeddings	F1 93-96%
FLAN-T5	HR Q&A	Instruction tuned LLM	Benchmark
Gemma	AI assistant	Enterprise reporting	Benchmark
Phi-3	Lightweight assistant	Efficient reasoning	Benchmark
Qwen	Multilingual analytics	Advanced reasoning	Benchmark
DeepSeek	Code & analytics	Strong reasoning	Benchmark

Conclusion

The newly proposed models broaden the analytical capabilities of the project beyond traditional machine learning. Ensemble models improve prediction accuracy, deep learning models capture complex workforce patterns, clustering algorithms identify talent groups, anomaly detection models discover unusual employee behavior, and modern large language models enable intelligent reporting and conversational analytics. Together with XGBoost, Random Forest, and LightGBM, these technologies create a scalable, explainable, and intelligent workforce analytics platform suitable for modern HR decision-making.